



Systematic Layout Planning (SLP)

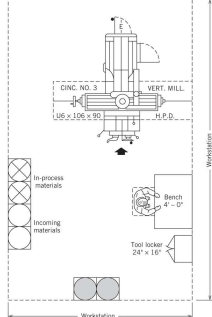
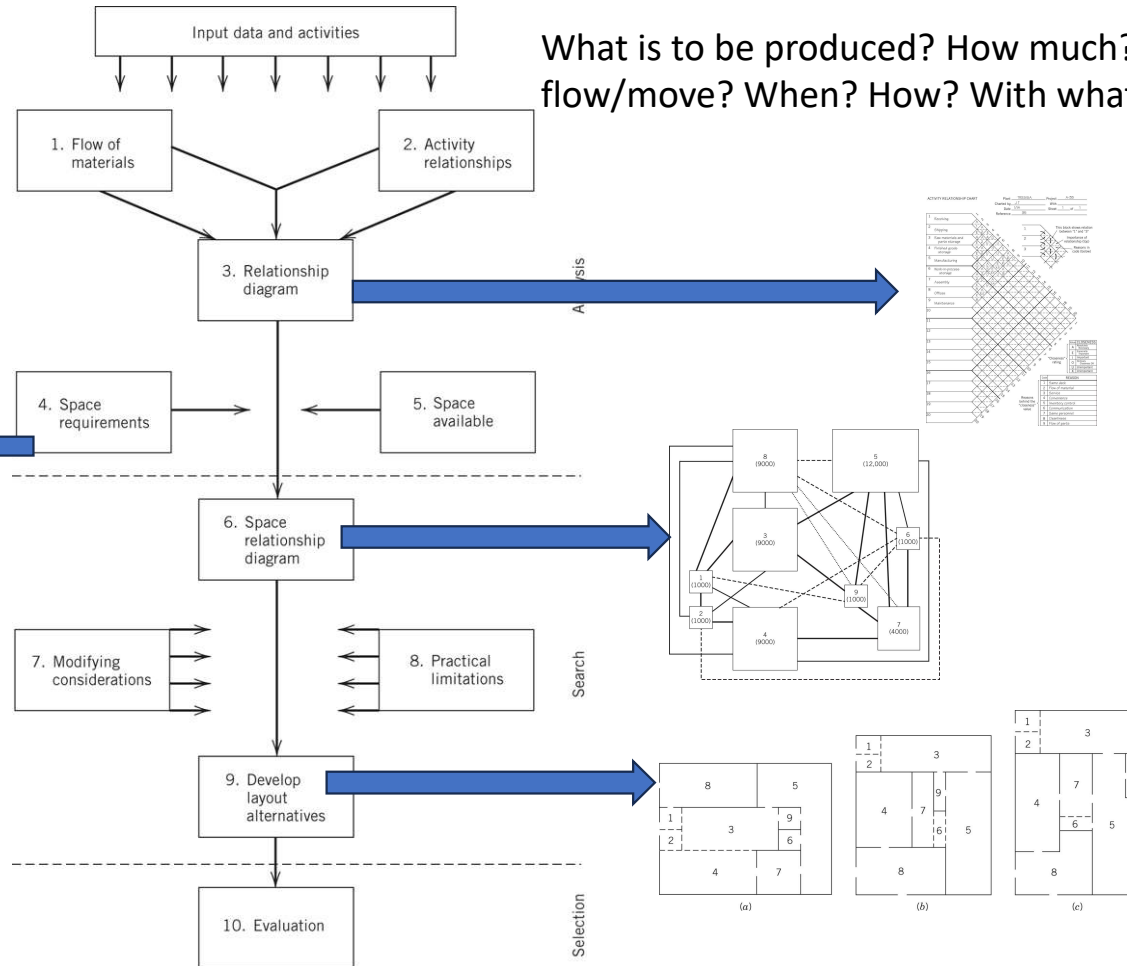
March 2025
W Kruger



SLP diagram logic

What is to be produced? How much? How will it flow/move? When? How? With what? = DATA gathered

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)	4x12=48x5=240	20x5=100	20x5=100	240+100+100=440
Screw m/c (6)	4x15=60x6=360	40x6=240	20x6=120	360+240+120=720
Chuckers (2)	6x5=30x2=60	50x2=100	20x2=40	60+100+40=200
Nett total				440+720+200=1360+desk=1364
Aisle %				177
TOTAL AREA				1364+177=1541sq m

Systematic Layout Planning for a Floor of a Department Store – example1 to illustrate SLP logic

A. Relationship Chart (based on Tables B and C)

From	To				Area (sq. ft.)
	2	3	4	5	
1. Credit department	I 6	U —	E 4	U —	100
2. Toy department		U —	I 1	A 1,6	400
3. Wine department			U —	X 1	300
4. Camera department				X 1	100
5. Candy department					100

Letter Closeness rating
Number Reason for rating

1 A Between where is A?
1 E
2 I
4 U
2 X Between where is X's?

B.

CODE	REASON*
1	Type of customer
2	Ease of supervision
3	Common personnel
4	Contact necessary
5	Share same space
6	Psychology

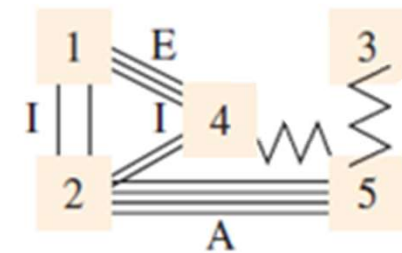
*Others may be used.

C.

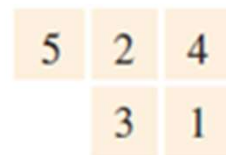
VALUE	CLOSENESS	LINE CODE*	NUMERICAL WEIGHTS
A	Absolutely necessary	=====	16
E	Especially important	=====	8
I	Important	=====	4
O	Ordinary closeness OK	=====	2
U	Unimportant	=====	0
X	Not allowed	^ ^ ^	-80

*Used for example purposes only.

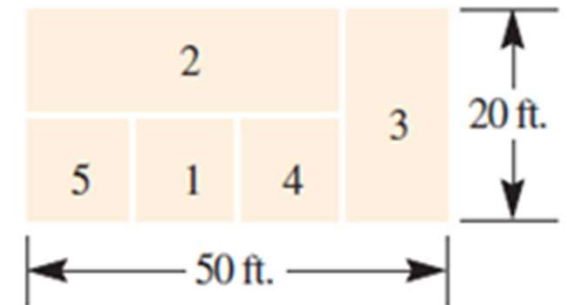
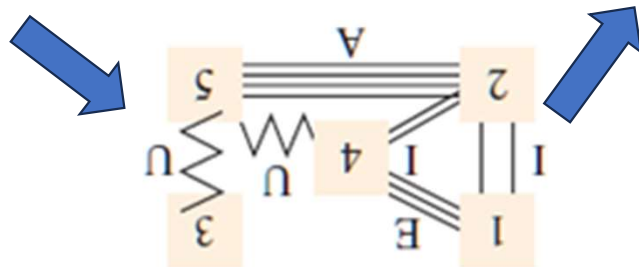
Systematic Layout Planning for a Floor of a Department Store – example to illustrate SLP logic



Initial relationship diagram
(based on Tables A and C)



Initial layout based on
relationship diagram
(ignoring space and
building constraints)

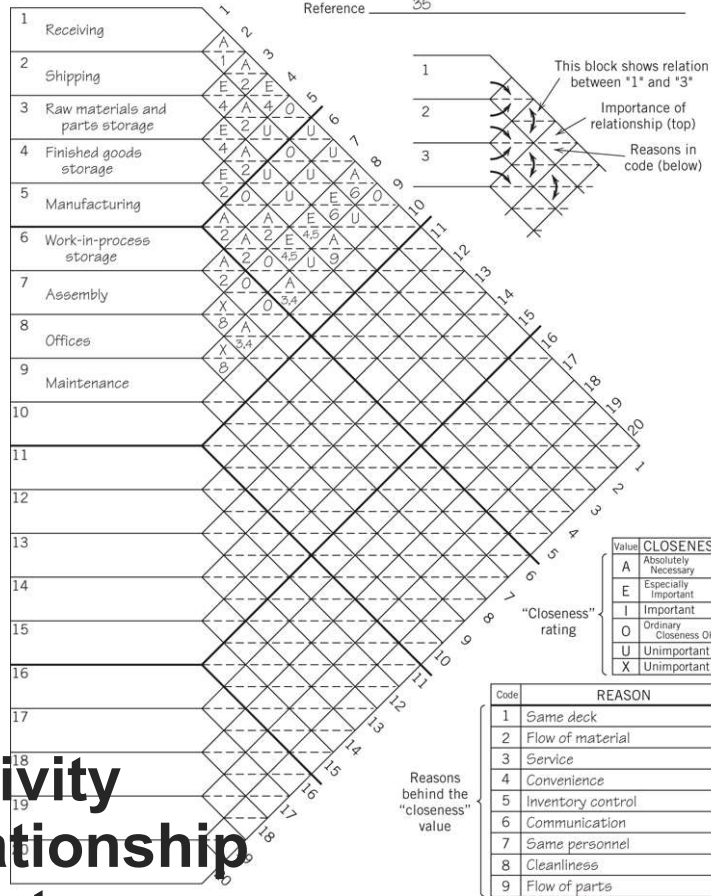


Final layout adjusted by square
footage and building size

Activity relationship chart

ACTIVITY RELATIONSHIP CHART

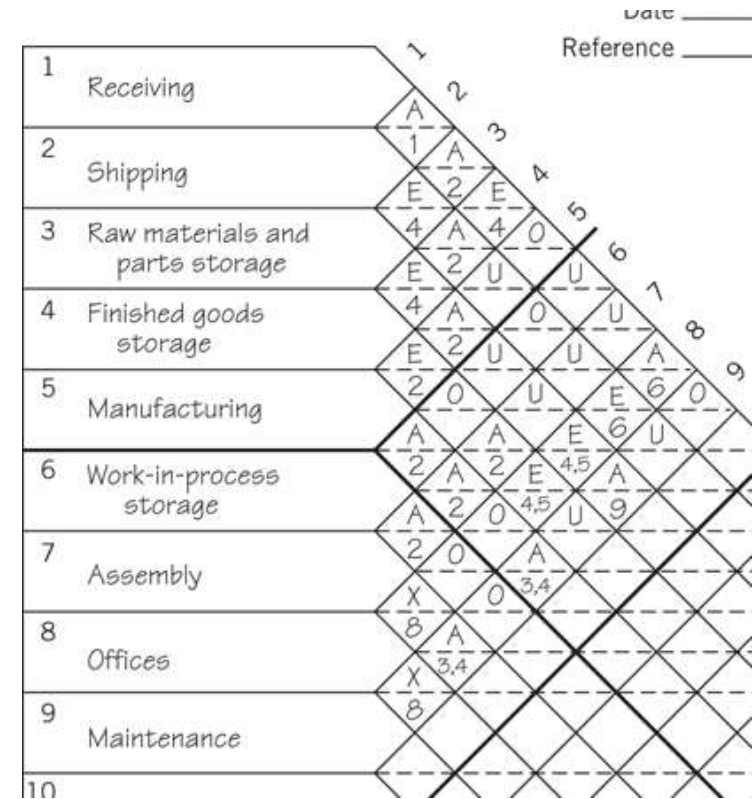
Plant TRESISA Project A-35
 Charted by J.T With _____
 Date 1/14 Sheet 1 of 1
 Reference 35



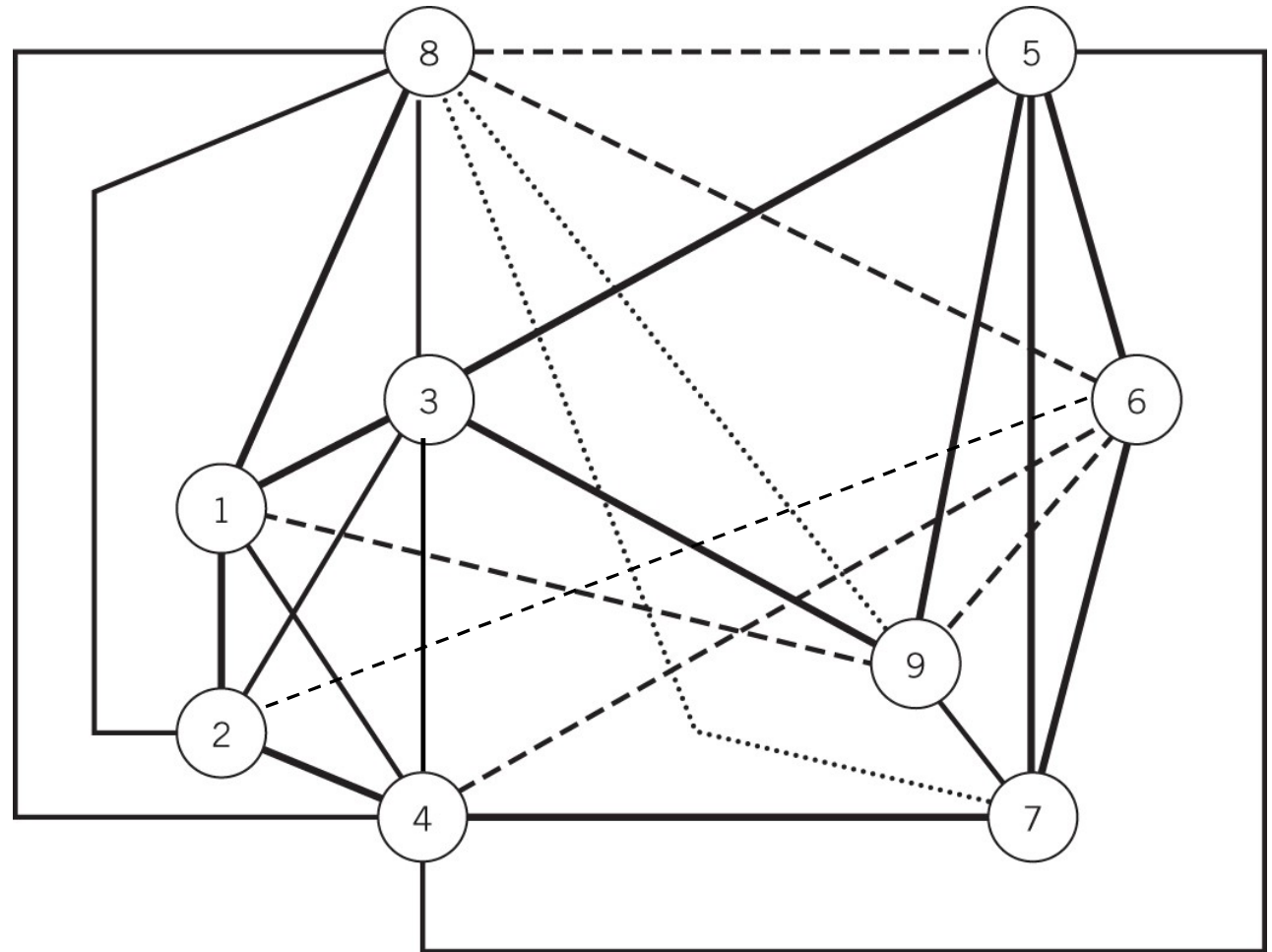
SLP – example2 p300 to illustrate SLP logic with activity relationship chart given

REASONS =
 Out of Scope

A – 10 **=====**
 E – 7 **=====**
 I – 0 **=====**
 O – 7 **-----**
 U – 8 **no line**
 X – 2 **.....**

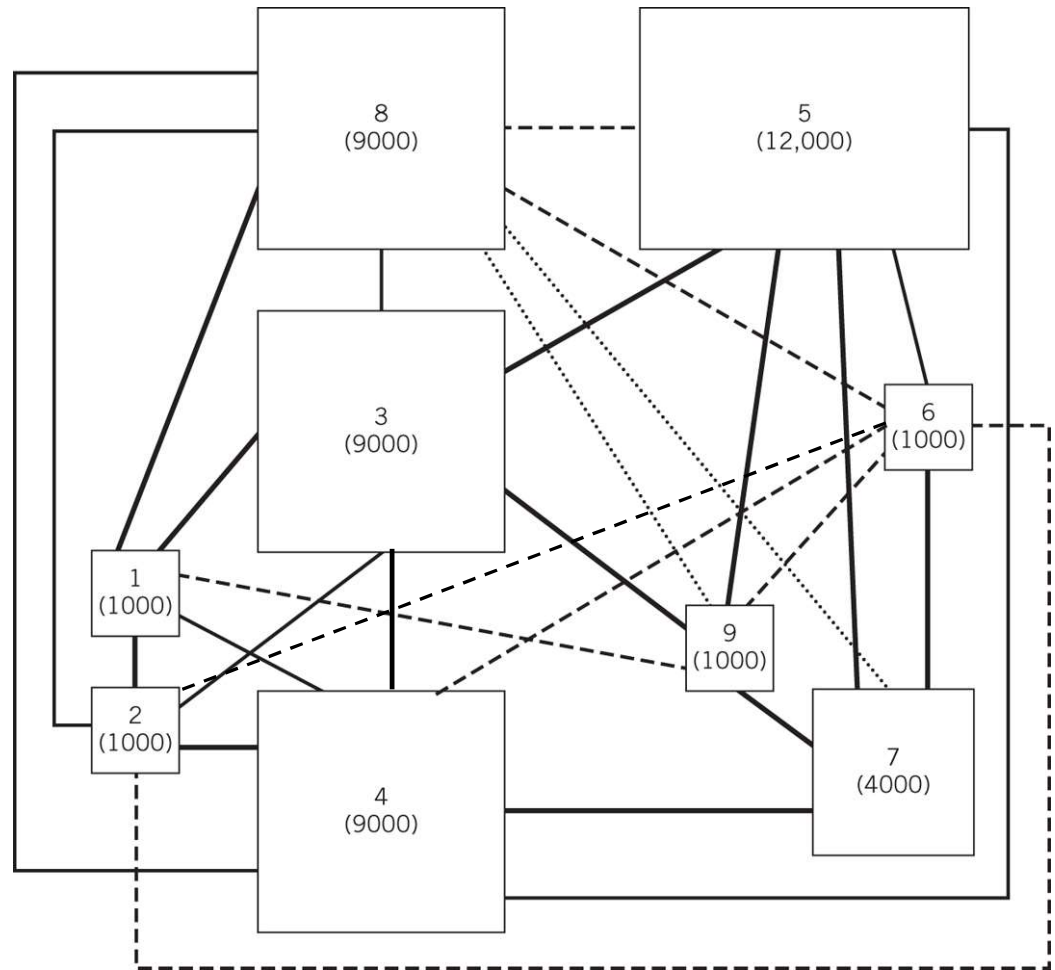


Resulting Relationship diagram

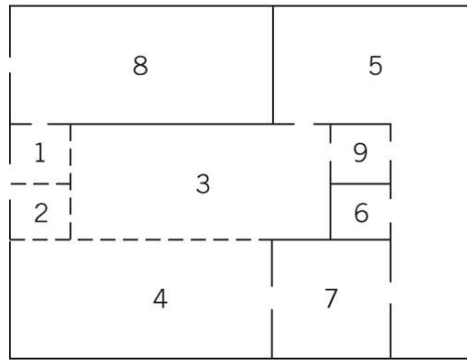


Resulting Space relationship diagram

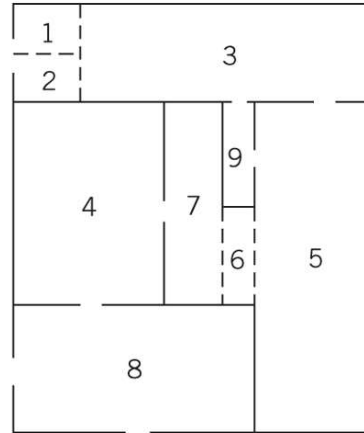
(imagine rubber
bands)



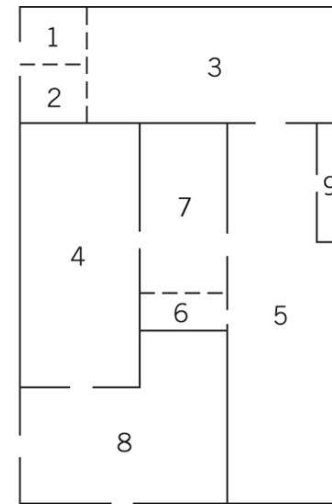
Resulting Alternative Block Layouts



(a)



(b)



(c)

fig_06_07

From-To chart

Department Name	Area (ft ²)	No. of Grids	Flow							
			A	B	C	D	E	F	G	H
1. A: Receiving	12,000	30	0	45	15	25	10	5	0	0
2. B: Milling	8000	20	0	0	0	30	25	15	0	0
3. C: Press	6000	15	0	0	0	0	5	10	0	0
4. D: Screw m/c	12,000	30	0	20	0	0	35	0	0	0
5. E: Assembly	8000	20	0	0	0	0	0	65	35	0
6. F: Plating	12,000	30	0	5	0	0	25	0	65	0
7. G: Shipping	12,000	30	0	0	0	0	0	0	0	0
8. H: Dummy	2000	5	0	0	0	0	0	0	0	0

table_06_03

Ranking (or given, gut feel, or sometimes calculation):

Max = 90 / 5 = intervals

0-18 = U

19-36 = O

37-54 = I

55-72 = E

73-90 = A

SLP – example 6.1 starting from From-To chart Ex 6.1-6.4

Constraints – Receiving(A) and Shipping(G) already placed next to each other, H = expansion space

Activity relationship chart

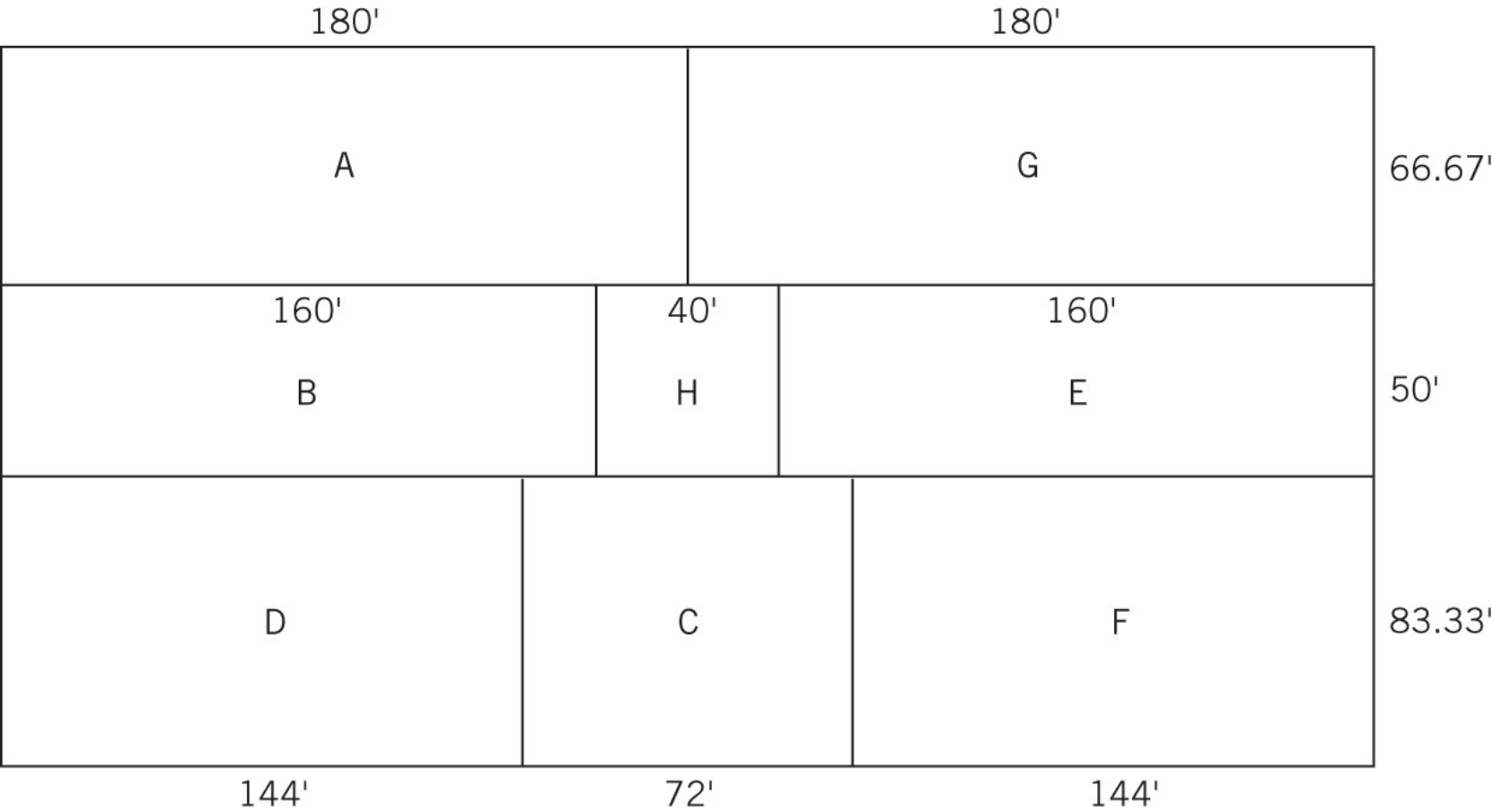
	A	B	C	D	E	F	G	H		A	B	C	D	E	F	G	H
A	0	45	15	25	10	5	0	0	A	—	I	U	O	U	U	U	U
B		0	0	50	25	20	0	0	B		—	U	I	O	O	U	U
C			0	0	5	10	0	0	C			—	U	U	U	U	U
D				0	35	0	0	0	D				—	O	U	U	U
E					0	90	35	0	E					—	A	O	U
F						0	65	0	F						—	E	U
G							0	0	G							—	U
H								0	H								—

(a) Flow-between chart

(b) Relationship chart

table_06_04

SLP – resulting block layout



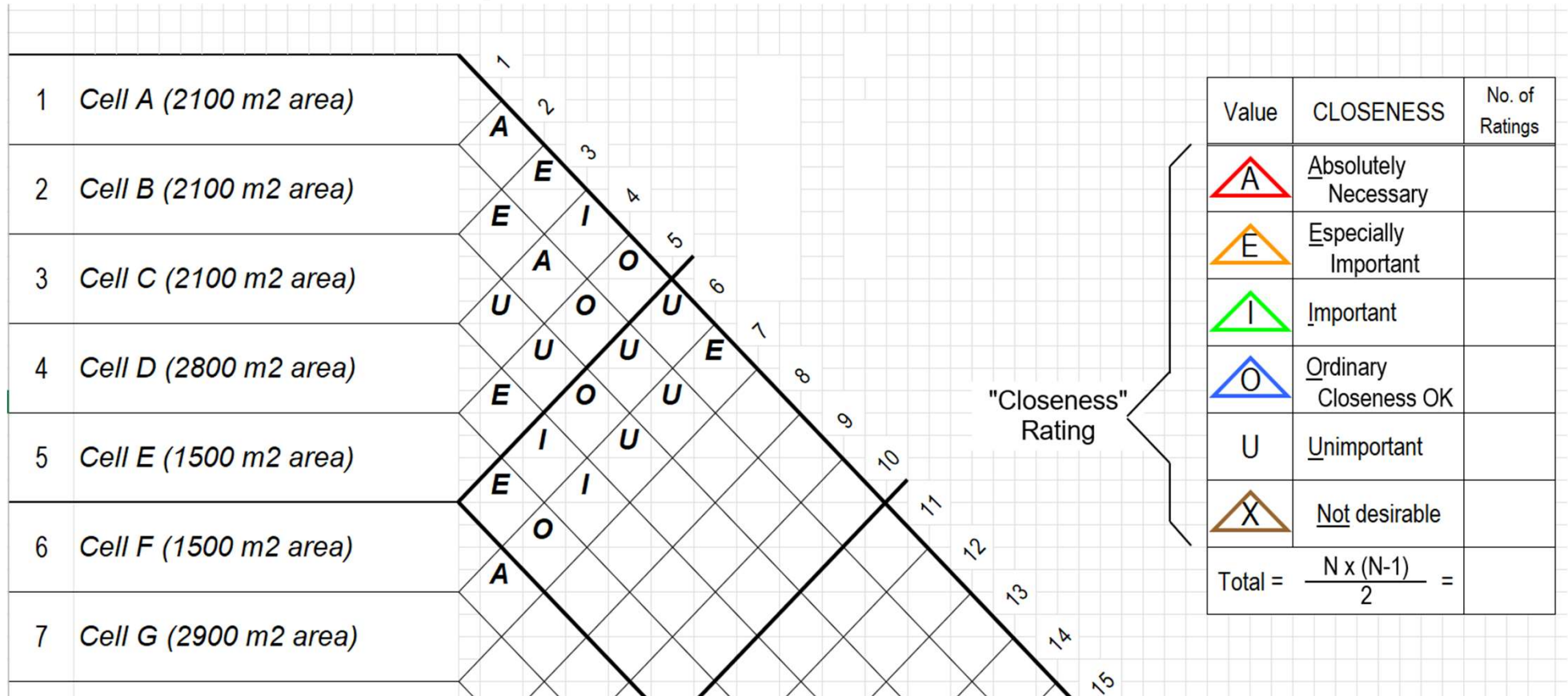
SLP – further examples....

Problems 6.10 in Tompkins book – 7 departments, different sizes

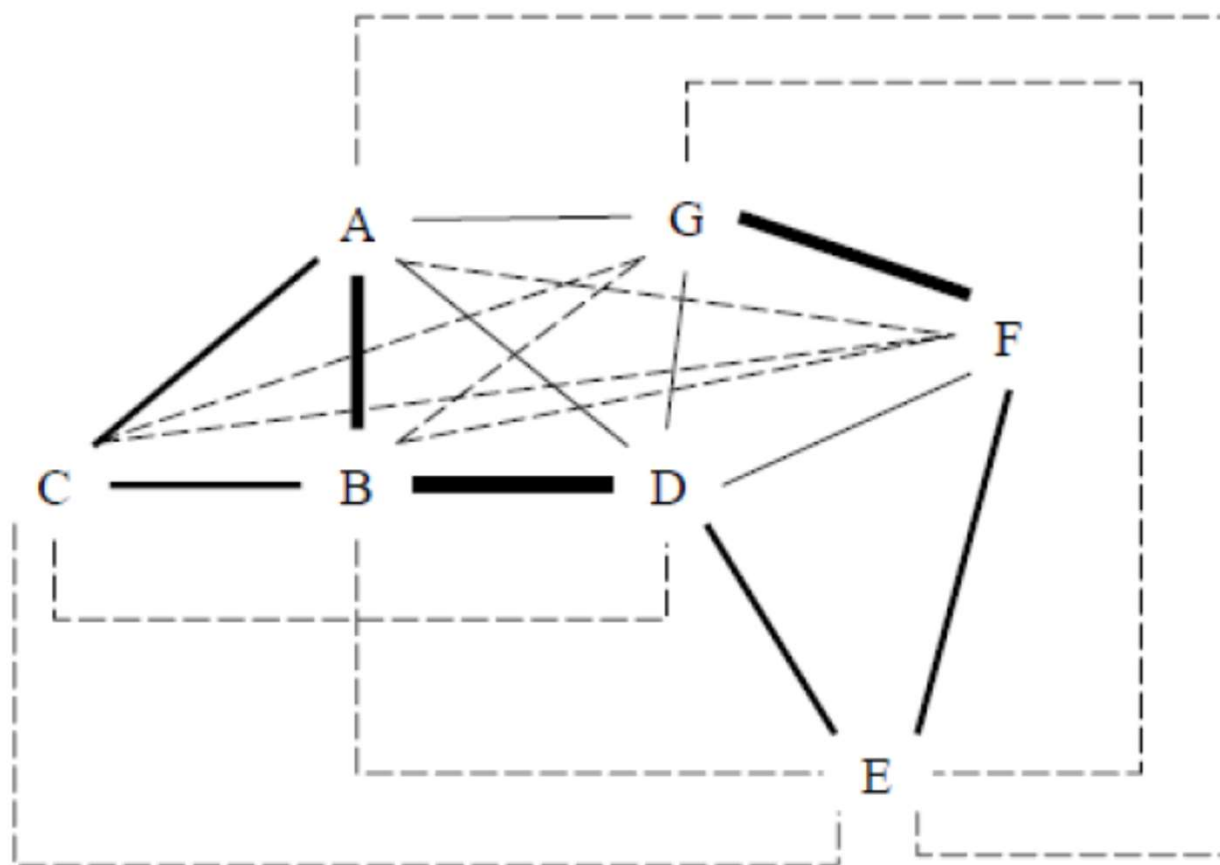
Chase textbook ch8 question 11 (pdf file) – 8 departments, same size, nr of lines is like rubberband

www.richardmuther.com for free working forms.(RMA130)

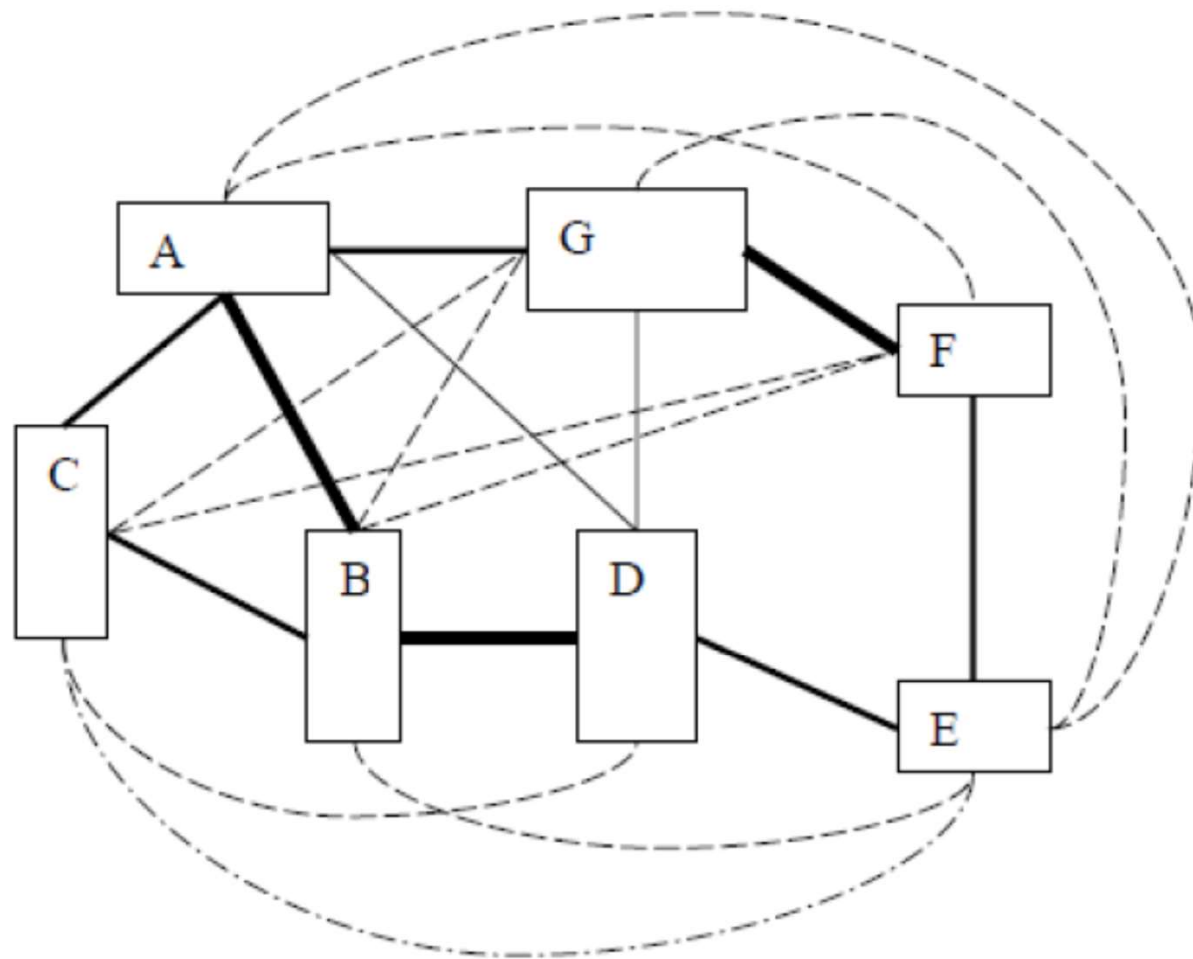
Problem 6.10 – 7 departments



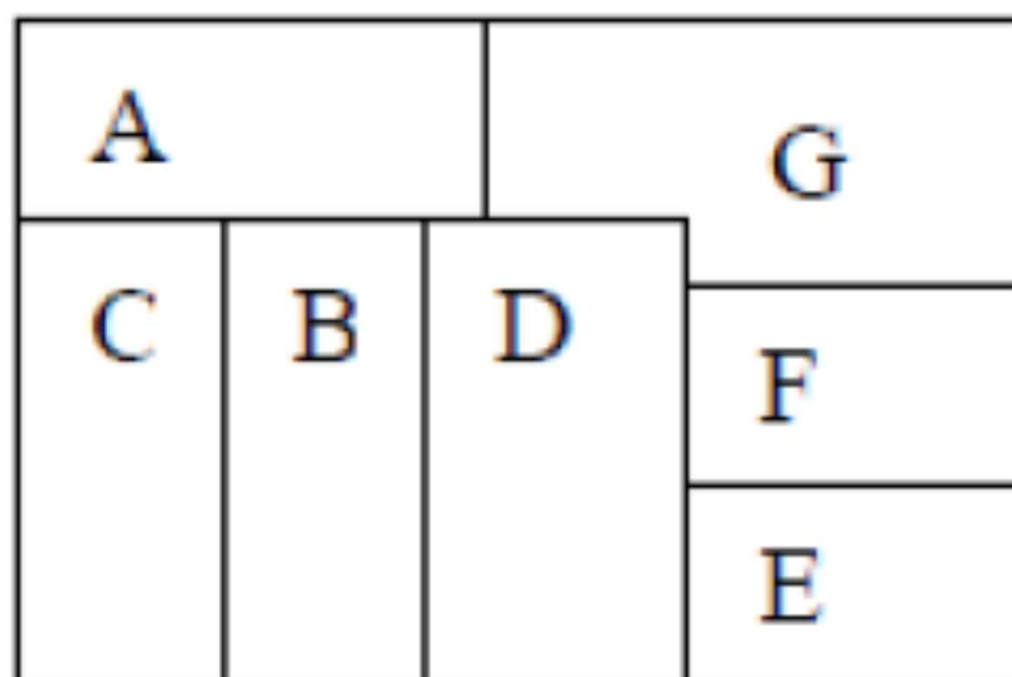
6.10a Relationship Diagram



6.10 Space Relationship Diagram



6.10b Block Layout using SLP



Other learning materials on SLP....

Muther himself: <https://www.youtube.com/watch?v=3orNXGODkis>

<https://www.youtube.com/watch?v=9xlA--aZoSM>

And again for the free forms....www.richardmuther.com

TUTORIAL :

6.8 with 6 departments starting at From-To chart

6.8a From-To Chart

Dept.	A	B	C	D	E	F
A	–	13,320	1,800	0	0	0
B	0	–	11,400	6,600	4,920	5,400
C	0	6,600	–	2,400	4,200	3,600
D	0	3,000	1,200	–	5,040	960
E	0	5,400	7,800	1,200	–	5,160
F	0	0	0	0	0	–

Flow-Between Chart

6.8

Dept.	A	B	C	D	E	F
A	–	13,320	1,800	0	0	0
B		–	18,000	9,600	10,320	5,400
C			–	3,600	12,000	3,600
D				–	6,240	960
E					–	5,160
F						–

Determine rankings....let us see what you do...

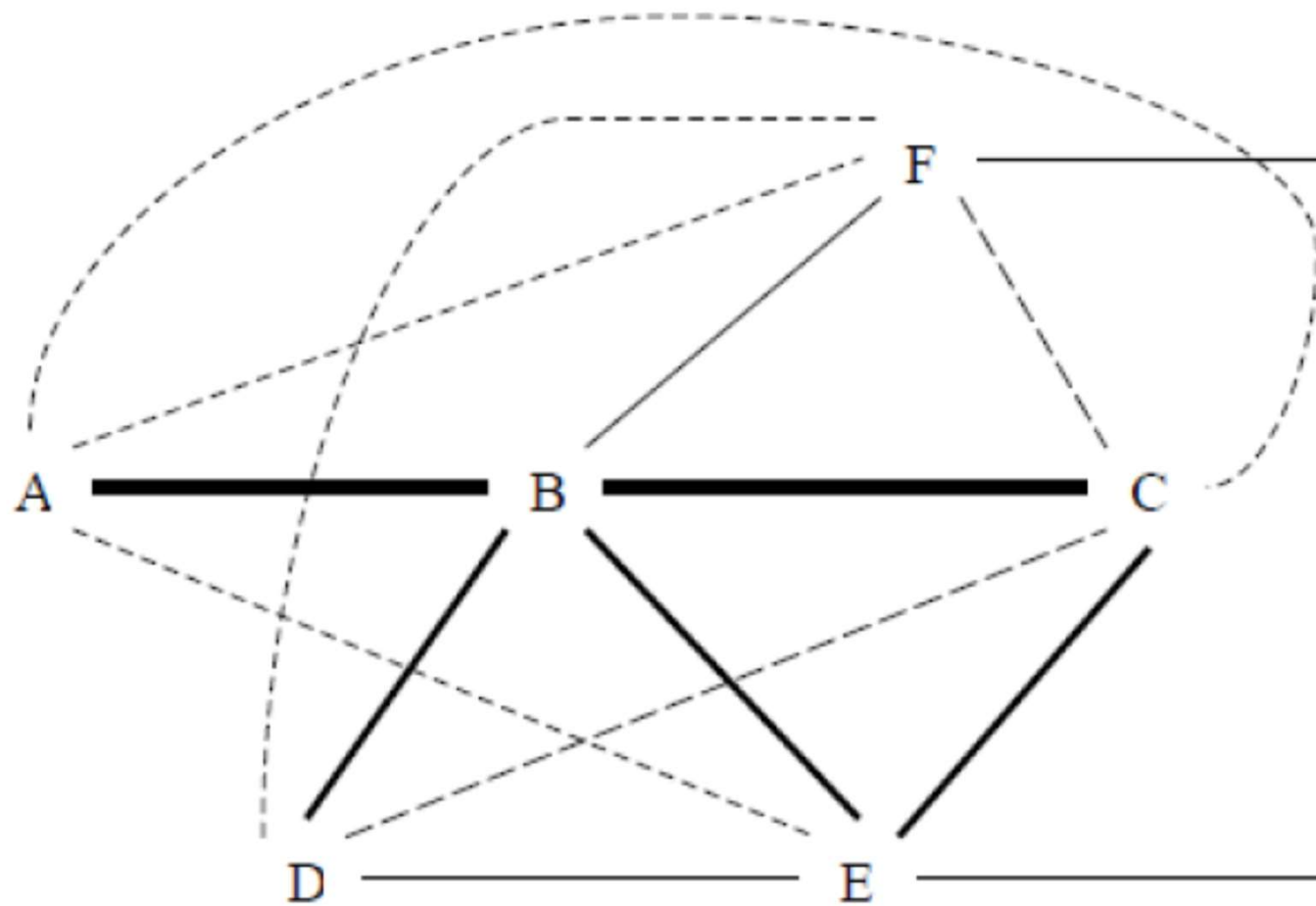
Activity Relationship Chart

6.8

Dept.	A	B	C	D	E	F
A	–	A	U	U	U	U
B		–	A	E	E	I
C			–	O	E	O
D				–	I	U
E					–	I
F						–

Relationship Diagram

6.8



6.8 Block Layout

